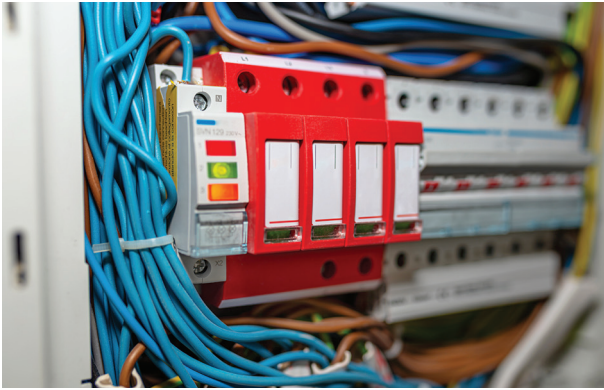




installers to bond every rack to a single grounding bar and to keep surge protectors under 0.5 Ω to earth.

Red flags checklist

- PoE switch at 90% budget during cold boot. Fix: split APs across two switches.
- UPS loaded past 80%. Fix: buy the next size up; APC derates above that.
- No bypass switch. Fix: ₹5 k wall-mounted breaker keeps you online during maintenance.

Hardware shopping list essentials

Below are three spend brackets you can take to any distributor or online cart. All prices are street averages in India, exclusive of GST; swap brands as long as you match the specs.

Spec Cheat-Sheet

- **Router:** look for dual WAN (one fibre, one 4G/5G), at least 900 Mbps NAT throughput, and IPSec/WireGuard hardware off-load. That keeps video calls smooth under load.
- **Switch:** a basic Layer 2 PoE model is fine until you start inter-VLAN routing on the switch itself; then move to Layer 3 Lite.



- **AP:** nearly every Wi-Fi 6E enterprise radio tops out at 22-23 W over PoE+ – plan your budget accordingly. A Cisco white paper warns that 6 GHz range drops ~15% indoors, so space APs every 10 m, not 12.
- **UPS:** pick online-double-conversion models if your area sees frequent brown-outs; LocalCircles found 38% of Indian homes still face daily interruptions.

Step-by-Step buying process

1. **List your loads:** PoE devices $\times$ max W = subtotal; add 20 % growth. Use FS.com's worksheet to double-check.
2. **Match switch budget:** Choose a PoE switch whose stated power budget $\geq$ subtotal $\times$ 1.2. Omnitron's guide shows the formula in plain English.
3. **Pick router to fit throughput:** Aim for VPN throughput $\geq$ 40 % of your ISP's advertised downlink to avoid bottlenecks when staff tunnel in.
4. **Size the UPS:** Total Watts $\times$ 0.25 = Wh for 15 min runtime; select a model meeting or exceeding that figure.
5. **Plan redundancy early:** If budget allows, buy two smaller UPS units rather than one big one; APC shows how "parallel redundant" beats single-unit uptime at modest extra cost.



India-specific procurement tips

- **BIS certification:** Import-only gear must carry a BIS R-number; customs now holds non-certified networking equipment.
- **GST input credit:** Routers and switches fall under HSN 8517 – claim the 18 % input tax credit against output GST.
- **Local warranty networks:** Brands like APC and Cisco run pan-India advance-replacement programmes – vital if your only switch dies during exam season at an ed-tech startup.

Key takeaways

1. **Capacity before brand:** Run the math – PoE, UPS, runtime – then pick logos.
2. **Redundancy saves rupees:** A bypass switch or second UPS costs less than an hour of payroll downtime.
3. **Specs, not hype:** Tri-band Wi-Fi 6E is pointless if your fibre link crawls at 100 Mbps; match layers.
4. **India's grid wobbles** – plan accordingly: Daily blips and heat-wave brown-outs aren't going away; design for them. With the blueprint, heat-map, PoE budget and hardware shortlist in hand, you're ready to mark cable routes on the office plan and place that first purchase order – confident that every watt, megabit and rupee has been accounted for. **d**